
FlowTeam

Frontend Design Guide

Version 1.0 • Next.js + Tailwind CSS + shadcn/ui

This document is the single source of truth for all UI decisions in FlowTeam.

[Colors](#) • [Typography](#) • [Spacing](#) • [Components](#) • [Patterns](#) • [Motion](#) • [Accessibility](#)

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1. Introduction

This design guide governs the visual language of FlowTeam — a team management platform built with Next.js 14, Tailwind CSS, and shadcn/ui. Every decision here has a reason. Follow the system rather than improvising to keep the product coherent as it scales across five phases of development.

Core philosophy

- Flat and clean. No gradients, shadows, noise textures, or decorative effects.
- Functional color. Color encodes meaning (priority, status, role) — never decoration.
- Consistent weight. Two font weights only: 400 regular and 500 medium. Never 600 or 700.
- Sentence case everywhere. Never Title Case or ALL CAPS in UI labels.
- Always skeleton, never empty containers. Every loading state has a designed appearance.
- Every error surfaces. Toast on every mutation failure, no silent failures.

When in doubt, refer to this document. If something is not covered here, match the closest existing pattern rather than inventing a new one.

2. Color system

FlowTeam uses a structured token-based color system built on Tailwind CSS custom properties. All colors adapt automatically to light and dark mode. Never hardcode hex values in components — always use the token.

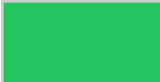
2.1 Brand primary — Indigo

Indigo is the single brand color. It appears on primary buttons, active sidebar items, focus rings, links, progress fills, and the app logo mark.

Swatch	Token name	Hex	Usage
	Indigo 50	#EEF2FF	Backgrounds, hover fills, badge bg
	Indigo 200	#C7D2FE	Borders on info elements
	Indigo 400	#818CF8	Accent icons, secondary actions
	Indigo 500	#6366F1	Primary buttons, active nav, links
	Indigo 600	#4F46E5	Button hover state
	Indigo 800	#3730A3	Button active/pressed state

2.2 Semantic colors

These map to Tailwind's semantic CSS variables and automatically adapt to dark mode via CSS custom properties.

Swatch	Semantic name	Hex (light)	Usage
	Success	#22C55E	Completed tasks, positive trends, done states
	Warning	#F59E0B	Due-today dates, at-risk health scores
	Danger	#EF4444	Overdue tasks, error toasts, delete actions
	Info	#6366F1	Informational badges, @mention highlights
	Muted	#94A3B8	Placeholder text, disabled states

2.3 Priority color system

Priority colors are fixed and never change. Users learn them once and rely on them everywhere in the product. They appear as left-border accents on task cards, in priority badge backgrounds, and in chart series.

Swatch	Priority	Border hex	Badge bg	Badge text
	Urgent	#EF4444	#FEE2E2	#991B1B
	High	#F97316	#FEF3C7	#92400E
	Normal	#3B82F6	#EFF6FF	#1E40AF
	Low	#94A3B8	#F1F5F9	#475569

2.4 Project palette (8 preset colors)

When creating a project, users pick from 8 curated colors. These are shown as left-border accents on project cards, event backgrounds on the calendar, and color dots in the sidebar. The palette is intentionally limited to prevent visual noise.

Indigo	Pink	Teal	Orange	Violet	Green	Yellow	Slate
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3. Typography

The default typeface is Inter (loaded via next/font). Fall back to the system sans-serif stack. All text uses only two weights: 400 and 500. Using bolder weights makes the UI feel heavy and inconsistent with shadcn/ui's defaults.

3.1 Type scale

Role	Size / Weight	Line height	Usage
Display	30px / 500	1.2	Auth page headings, onboarding
H1	24px / 500	1.3	Page titles
H2	20px / 500	1.3	Section titles within a page
H3	16px / 500	1.4	Card headings, panel titles
Body	14px / 400	1.7	Descriptions, task text, prose
Body small	13px / 400	1.5	Secondary info, metadata
Caption	12px / 400	1.4	Timestamps, helper text
Label	11px / 500	1.2	Section headers (uppercase), badges
Mono	12px / 400	1.5	IDs, hex values, code snippets

3.2 Typography rules

- Sentence case everywhere. Labels, headings, buttons, nav items — no exceptions.
- Maximum line length: 65ch for body text. Prevents unreadable long lines.
- Truncate overflowing text with ellipsis. Never let UI text wrap unexpectedly.
- Mono font (font-mono) for IDs, hex values, timestamps in audit log, and code.
- Never use font-weight 600 or 700 — override shadcn defaults where needed.
- Heading colors use text-primary (auto dark mode). Never hardcode heading colors.

4. Spacing & layout

4.1 Spacing scale

Use only values from this scale. Do not use arbitrary values like 10px or 15px.

Token	Value	Common usage
space-1	4px	Icon-to-text gap, tight inline spacing
space-2	8px	Component internal padding, avatar group gap
space-3	12px	Card grid gap, form field gap
space-4	16px	Card padding (compact), section gap
space-5	20px	Card padding (comfortable)
space-6	24px	Between sections within a page
space-8	32px	Between major page sections
space-12	48px	Page-level vertical rhythm

4.2 Border radius

Token	Value	Usage
rounded-sm	4px	Inline code chips, small tags
rounded-md	8px	Buttons, inputs, small elements
rounded-lg	12px	Cards, panels, dropdowns, modals
rounded-xl	16px	Large dialogs, image containers
rounded-full	9999px	Avatars, badges, pills, status dots

4.3 App shell dimensions

Element	Size	Notes
Sidebar (expanded)	240px	Fixed, collapsible to 64px icon-only on mobile
Sidebar (collapsed)	64px	Icons only with tooltips on hover
Top header bar	48px	Breadcrumb + search + notifications
Content max-width	1280px	Centered, fluid below
Page padding	24px sides	24px top on all content pages

Element	Size	Notes
Kanban column width	280px fixed	Horizontal scroll container
Task detail slide-over	480px	Right-anchored, overlays content
Search modal	640px	Centered dialog, full-screen overlay

4.4 Responsive breakpoints

Breakpoint	Width	Layout changes
sm	640px	Mobile landscape — single column, stacked nav
md	768px	Tablet — 2-col grids, drawer sidebar
lg	1024px	Sidebar visible, 3-col project grid
xl	1280px	Full layout, 4 stat cards in a row
2xl	1536px	Wide screens — cap max-width at 1280px

4.5 Grid patterns

Page / Section	Grid definition
Stat cards (dashboard)	grid-cols-4 → grid-cols-2 (md) → grid-cols-1 (sm)
Project cards	grid-cols-3 → grid-cols-2 (md) → grid-cols-1 (sm)
Dashboard main	2/3 + 1/3 split (task list + activity feed)
Kanban board	flex row, overflow-x-auto, columns 280px fixed
Settings page	Full width, max-w-3xl, single column
Reports page	Full width with tab navigation

Always use `minmax(0, 1fr)` in CSS grids to prevent children from pushing columns past the container.

5. Component reference

5.1 Badges & pills

Badges encode fixed meaning. Never repurpose a priority color for something other than priority. Role badges use separate indigo/teal/gray palette to avoid confusion with priority.

Priority badges

Urgent	High	Normal	Low
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Status badges

To do	In progress	In review	Done
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Role badges

Admin	Member	Viewer
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5.2 Avatars

Avatar color is assigned deterministically by hashing the user's ID, ensuring each user always gets the same color across all sessions and devices. Use initials (first + last name initial). Available avatar color pairs:

AK	SR	JL	MP	TR
20px — list	24px — card	32px — detail	40px — profile	Stack: -8px overlap

5.3 Task cards

The task card is the most common UI element. It appears in kanban columns, my-tasks lists, search results, and calendar events. The left border always encodes priority using the fixed priority color system.

Implement JWT refresh token rotation

Urgent Backend Overdue — was due Jan 12 AK

Write unit tests for task reorder endpoint

Normal Frontend Jan 20 □ 2/5 subtasks JL

Task card anatomy:

- Left border (3px): fixed priority color — never use for anything else
- Title: 14px / 500 — truncate at 2 lines (line-clamp-2)
- Metadata row: priority badge + label chips + due date + assignee avatar
- Due date: green if future, amber if today, red if overdue
- Assignee avatar: right-aligned, 20px, initials
- Drag handle: opacity-0 group-hover:opacity-100, appears on left side on hover
- Hover: show 'Open' icon button on the right side

5.4 Stat (metric) cards

Used on the dashboard for at-a-glance numbers. These are surface-colored (bg-muted), never white with a border. The value is always the most prominent element.

Open tasks 24 <i>+3 this week</i>	Overdue 5 <i>+2 since yesterday</i>	Completed 18 <i>+22% vs last week</i>	Due today 3 <i>across 2 projects</i>
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5.5 Progress bars

6px height. Border radius 3px (full-round). Color encodes state: indigo for normal progress, green for complete, amber for at-risk, red for over-estimate.

Always show the percentage label alongside the bar. Never rely on bar width alone to communicate the value.

5.6 Buttons

Use shadcn/ui Button component. Three variants are used in FlowTeam:

Variant	Tailwind class	Usage
Default	bg-primary text-white	Primary actions: Save, Create, Submit
Outline	border bg-transparent	Secondary actions: Cancel, Export, Filter
Ghost	no border, hover bg-muted	Toolbar icons, navigation items
Destructive	bg-destructive text-white	Delete, remove — always confirm first

- Loading state: replace text with spinner + 'Saving...' text, disable the button

- Success state: brief 'Saved ✓' text for 2 seconds then revert — only on explicit save actions
- Active scale: transform: scale(0.97) on active/pressed — subtle tactile feedback
- Icon buttons: always include tooltip (title attribute) for accessibility

6. UI patterns

6.1 Do / don't

Do	Don't
✓ 0.5px borders on all cards and inputs	✗ Gradients or box-shadows
✓ Sentence case in all UI labels	✗ Font-weight 600 or 700
✓ Skeleton loaders while data fetches	✗ ALL CAPS or Title Case labels
✓ Debounce search inputs at 300ms	✗ Empty containers without empty state
✓ Optimistic updates on every mutation	✗ Unthrottled API calls on keystroke
✓ Toast notification on every error	✗ Hardcoded color hex in components
✓ Truncate overflow text with ellipsis	✗ Spinners without timeout/error handling
✓ Empty state for every empty list	✗ Tables wider than their container
✓ minmax(0, 1fr) in all CSS grids	✗ position: fixed inside modals

6.2 Empty states

Every list, table, and board must have a designed empty state. Never show an empty container. The empty state should be contextual — it should tell the user why it's empty and what to do.

Location	Heading	CTA
Kanban column	"Drop tasks here"	None — drag target
My tasks — overdue	"No overdue tasks"	None — positive state
Projects list	"No projects yet"	"Create your first project"
Search results	"No results for '...'"	"Try different keywords"
Notifications	"You're all caught up"	None — positive state
Audit log	"No events match filters"	"Clear filters"
Calendar	"No tasks due this month"	"Add a due date to a task"

6.3 Toast notifications

Use sonner for all toasts. Position: bottom-right. Duration: 4 seconds (errors persist until dismissed). Never stack more than 3 toasts.

Type	Trigger	Message pattern	Action
Success	Task created	"Task created"	"Open" → opens detail
Success	Task moved	"Moved to {column name}"	None
Success	Export ready	"Export ready — {filename}"	"Download"
Error	Any mutation fail	"Failed to {action}. Try again."	None
Info	File uploaded	"File uploaded"	None
Warning	Rate limit hit	"Slow down — too many requests"	None

6.4 Loading states

Every async operation must have a visible loading state. Use skeleton components — not spinners — for initial page loads. Use inline spinners on buttons and refresh indicators. Never show a blank screen.

Context	Loading pattern
Page initial load	Full skeleton matching the layout shape
Button mutation	Spinner replaces icon, text changes to 'Saving...'
Search input	Spinner in right side of input while API call fires
Task detail open	Skeleton for each section (properties, subtasks, attachments)
Calendar month change	keepPreviousData: true — old month visible until new loads
Infinite scroll	3 skeleton cards at bottom while loading next page

6.5 Optimistic updates

Every mutation that changes visible state must use an optimistic update. The pattern is always: snapshot → update cache → API call → on error: rollback + toast.

Mutation	Optimistic behavior
Drag task to column	Card moves instantly, API call fires after
Change task priority	Badge updates instantly in card and detail panel
Toggle subtask complete	Checkbox ticks, progress bar updates instantly
Mark notification read	Blue dot disappears, unread count decrements
Send chat message	Message appears in list immediately, greyed until confirmed
Change task assignee	Avatar updates in card instantly

Always capture the snapshot BEFORE the optimistic update fires, not after. This is the most common mistake when implementing onMutate in React Query.

7. Motion & animation

Motion in FlowTeam is functional — it reinforces state changes and guides attention. Never animate for decoration. Keep durations short. No skeleton shimmer animations.

7.1 Duration scale

Name	Duration	Usage
instant	0ms	Checkbox toggle, button active state
fast	100ms	Hover background, badge change
normal	150ms	Dropdown open, color change
medium	200ms	Modal backdrop, tooltip fade
slow	300ms	Panel slide-in, drawer open
progress	400ms	Progress bar fill on page load

7.2 Easing functions

Name	Value	Usage
ease-out	cubic-bezier(0, 0, 0.2, 1)	Elements entering the screen
ease-in	cubic-bezier(0.4, 0, 1, 1)	Elements leaving the screen
spring	cubic-bezier(0.34, 1.56, 0.64, 1)	Drag-drop release, modal pop-in
linear	linear	Spinners, progress bars, shimmer

7.3 Specific transitions

- Sidebar collapse: width 240px → 64px, duration 200ms, ease-in-out
- Task detail slide-over: $\text{translateX}(100\%) \rightarrow \text{translateX}(0)$, 300ms ease-out
- Modal enter: $\text{opacity } 0 + \text{scale}(0.97) \rightarrow \text{opacity } 1 + \text{scale}(1)$, 200ms ease-out
- Modal backdrop: $\text{opacity } 0 \rightarrow 0.45$, 200ms linear
- Notification dropdown: $\text{opacity } 0 + \text{translateY}(-4\text{px}) \rightarrow \text{visible}$, 150ms ease-out
- Drag card: lifted state adds slight $\text{scale}(1.02)$ and reduces opacity to 0.9
- Button press: $\text{scale}(0.97)$, 100ms, instant return
- Card hover: border-color transition, 150ms linear — no scale or shadow
- Skeleton loaders: static gray fills only — no shimmer or pulse animations
- Progress bar: fill width transition on mount, 400ms linear with 100ms delay per card

8. Accessibility

FlowTeam targets WCAG 2.1 AA compliance. These are the minimum requirements — not aspirational goals.

8.1 Color contrast

- All body text: minimum 4.5:1 contrast ratio against background
- Large text (18px+ bold or 24px+): minimum 3:1 ratio
- Never use color as the only means of conveying information (e.g. priority bars always have a badge too)
- Test every badge combination in dark mode — light-mode passing doesn't guarantee dark mode passes

8.2 Keyboard navigation

- All interactive elements reachable by Tab key
- Visible focus ring on every focused element: box-shadow: 0 0 0 2px #6366F1
- Escape closes any open modal, panel, dropdown, or tooltip
- Kanban board: tasks navigable with arrow keys when focused on a card
- Search modal: full arrow-key navigation through results, Enter to activate
- Dropdowns: arrow keys navigate options, Enter selects, Escape closes

8.3 Screen reader support

- All icon-only buttons must have aria-label
- Status changes (task moved, notification received) must use aria-live regions
- Skeleton loaders: use aria-busy='true' on the container
- Modal dialogs: use role='dialog', aria-modal='true', trap focus inside
- Progress bars: use role='progressbar', aria-valuenow, aria-valuemin, aria-valuemax
- Avatar stacks: expose full list via aria-label (e.g. 'Alex, Sara, and 3 more')

9. Keyboard shortcuts

All shortcuts use the global `useEffect` listener implemented in Phase 5. The shortcut help modal (Cmd+/) shows the full list. On Windows/Linux, substitute Ctrl for Cmd.

9.1 Navigation

Shortcut	Action
Cmd + K	Open global search modal
G then D	Navigate to /dashboard
G then P	Navigate to /projects
G then C	Navigate to /calendar
G then M	Navigate to /messages
Escape	Close any open modal, panel, or dropdown

9.2 Tasks

Shortcut	Action
Cmd + N	New task (when on a project board)
E	Edit task title (when task card focused)
Space	Toggle subtask complete (when subtask focused)
Cmd + Delete	Delete selected task (with confirm)

9.3 Messaging

Shortcut	Action
Enter	Send message
Shift + Enter	New line in message composer
Up Arrow	Edit last sent message
@	Trigger @mention autocomplete

9.4 General

Shortcut	Action
Cmd + /	Show keyboard shortcuts help modal
Cmd + Z	Undo last action (where supported)
Cmd + S	Save / confirm current form

10. Dark mode

Dark mode is mandatory. Every component must work in both modes. FlowTeam uses the Tailwind class strategy (class='dark' on <html>) toggled by the user's system preference or a manual toggle in settings.

10.1 Core rules

- Never hardcode hex colors in Tailwind classes — always use semantic tokens (text-primary, bg-card, border-border, etc.)
- Test every badge, card, and chart in dark mode before shipping
- Chart colors: use the same hex but reduce opacity to 0.85 in dark mode via CSS variable
- Code blocks: dark background (#1E1B4B) is the same in both modes — no inversion
- Indigo brand color inverts: in dark mode use indigo-400 (#818CF8) instead of indigo-500
- Avatar initials must be legible: test all 5 color pairs in dark mode
- Skeleton loaders: use bg-muted which auto-adapts to dark mode

10.2 Shadcn/ui dark mode token mapping

Token	Light mode value	Dark mode value
background	#FFFFFF	#0F172A
foreground	#0F172A	#F8F AFC
card	#FFFFFF	#1E293B
muted	#F1F5F9	#1E293B
muted-foreground	#64748B	#94A3B8
border	#E2E8F0	#334155
primary	#6366F1	#818CF8
primary-foreground	#FFFFFF	#0F172A

11. File & folder conventions

11.1 Component naming

- PascalCase for component files: TaskCard.tsx, ProjectCard.tsx
- camelCase for hooks: useTaskDetail.ts, useWebSocket.ts
- kebab-case for page folders: /task-detail/, /project-reports/
- Colocate component, types, and hook in same folder for complex components
- Index files (index.ts) for clean imports from component folders

11.2 Component folder structure

```
components/  
  ui/           ← shadcn generated components (never edit)  
  layout/       ← Sidebar, Header, AppShell  
  auth/         ← LoginForm, RegisterForm, OnboardingFlow  
  board/        ← KanbanBoard, Column, TaskCard, TaskDetail  
  dashboard/    ← StatCard, ActivityFeed, WorkloadChart  
  messaging/    ← ChatWindow, MessageBubble, Composer  
  calendar/     ← CalendarView, EventCard  
  search/       ← SearchModal, SearchResult  
  notifications/ ← NotificationBell, NotificationPanel  
  reports/      ← VelocityChart, BurndownChart, MemberTable  
  shared/       ← Avatar, Badge, PriorityBadge, EmptyState
```

11.3 Naming conventions

Thing	Convention	Example
Component files	PascalCase.tsx	TaskCard.tsx
Hook files	camelCase.ts	useTaskDetail.ts
Store files	camelCase.ts	boardStore.ts
Type files	camelCase.ts	task.types.ts
Page folders	kebab-case	/project-reports/
API files	camelCase.ts	tasksApi.ts
Constant files	SCREAMING_SNAKE	PRIORITY_COLORS.ts

12. Pre-ship design checklist

Run this checklist before marking any feature as ready for review. Each item must pass in both light and dark mode, on desktop and mobile.

12.1 Visual quality

- No hardcoded hex values — all colors use tokens
- No font-weight 600 or 700 anywhere in the component
- All text in sentence case (no Title Case, no ALL CAPS)
- All text truncates gracefully with ellipsis — test with long strings
- Empty state exists for every list, table, or card container
- Loading skeleton matches the shape of the loaded content
- No raw emoji in UI — only in intentional empty states or system messages

12.2 Interaction quality

- Every mutation shows a loading state on the triggering button
- Every mutation failure shows a toast with a clear error message
- Optimistic update fires before the API response
- Keyboard shortcut (Escape) closes any open panel, modal, or dropdown
- Tab order is logical — test by tabbing through the entire component
- All icon-only buttons have aria-label

12.3 Responsive

- Tested at 375px (iPhone SE), 768px (tablet), 1280px (desktop)
- No horizontal scroll on mobile (except intentional, like kanban)
- Sidebar collapses to drawer on mobile — test hamburger open/close
- Tables don't overflow their container — use table-layout: fixed if needed
- Touch targets are at least 44px × 44px on mobile

12.4 Dark mode

- All text is readable at sufficient contrast in dark mode
- All badges legible in dark mode — test all priority + role + status variants
- Charts and data visualizations have sufficient contrast in dark mode
- No pure white (#FFFFFF) backgrounds — use bg-card or bg-background tokens

FlowTeam Frontend Design Guide — v1.0
Keep this document updated as the design system evolves.